

# MOHIT KUMAR

Sikar, Rajasthan | [+91 9950147038](tel:+919950147038) | [mohitkumar4922251@gmail.com](mailto:mohitkumar4922251@gmail.com) | [Portfolio](#) | [LinkedIn](#) | [GitHub](#)

## SUMMARY

Backend-focused Software Engineer and **IIT Jodhpur Computer Science graduate (2026)** with experience building **scalable backend systems, REST APIs**. Skilled in **Python, Java, FastAPI, Spring Boot, PostgreSQL, Redis, Docker, and AWS**, with hands-on experience in **concurrency, idempotency, rate limiting, caching**. Co-authored research published at **ACM ICDCN 2026**. Seeking Backend Software Engineer and Software Development Engineer opportunities.

## EDUCATION

**Indian Institute of Technology (IIT), Jodhpur**

**Nov 2022 – Jun 2026**

B.Tech – Computer Science and Engineering

Jodhpur, Rajasthan

## TECHNICAL SKILLS

**Languages:** Java, Python, C++, TypeScript, SQL

**Backend & Web:** Spring Boot, FastAPI, Node.js, React, Next.js, REST APIs, Microservice

**Cloud & Databases:** Docker, AWS (EC2, S3, ECS, Lambda), PostgreSQL, MySQL, MongoDB, Redis

**Core CS & Tools:** DSA, OOP, Operating Systems, Computer Networks, DBMS, Distributed Systems, Git, GitHub

Actions, Prometheus, Grafana, Postman

## PROJECTS

**Payment Idempotency Proxy** | [GitHub](#)

**May 2026 – Jun 2026**

- Designed an idempotent payment proxy (FastAPI, Redis, PostgreSQL) to eliminate duplicate charges in concurrent flows, using Redis distributed locks (SET NX + Lua-based safe release) scoped by user, endpoint, and idempotency key
- Implemented SHA-256 payload fingerprinting to detect and reject idempotency-key reuse across mismatched request bodies, with cached final-response replay and tamper-evident audit logging.
- Containerized with Docker Compose; added Prometheus/Grafana observability and configurable retry/backoff; 29 unit and integration tests covering concurrent idempotency, locking behavior, and cross-key isolation.

**URL Shortener Service** | [GitHub](#)

**Aug 2025 – Oct 2025**

- Built for viral traffic resilience: Snowflake ID generation + Base62 encoding for compact links; cache-aside architecture with Redis read-through (1-hour TTL) and double-checked locking to prevent cache stampedes.
- Per-IP rate limiting (SlowAPI) and fully async request handling via SQLAlchemy 2.0 + asyncpg; stack: FastAPI, Redis, PostgreSQL.

**EdgeCraft CDN** | [GitHub](#)

**Jul 2026 – Aug 2026**

- Built a distributed CDN simulator in **C++17** modeling **LRU caching**, TTL-based expiry, and origin fallback; achieved up to **99.1% cache efficiency** under **Zipfian traffic**, exploiting real-world access skew to keep origin fetches down to 1% of requests.
- Implemented **consistent hashing with virtual nodes** for request routing across edge servers, minimizing key redistribution on node changes to only the joining/leaving node's share of keys, while keeping per-edge cache load balanced (**98–99%**) even after simulated node failures.
- Benchmarked across **100,000 requests** and **1,000 files** under Zipfian traffic distributions; reached **606K+ req/sec** throughput (**zero-latency benchmark**) and **99% origin offload**, tracking per-edge hit rate and load distribution throughout.

## EXPERIENCE

**Undergraduate Researcher**

**May 2025 – Aug 2025**

**Ubisys Lab, IIT Jodhpur**

Jodhpur, Rajasthan

- Co-authored **FedMeet** (ACM ICDCN 2026), a privacy-preserving federated learning framework for multi-sensor human activity recognition under non-IID settings using **Python, TensorFlow, and Flask**.
- Developed data preprocessing and training pipelines with sensor synchronization, Gaussian masking, and BiLSTM-based federated learning; achieved **87.97%** test accuracy, outperforming state-of-the-art baselines by **12%**.
- Evaluated adaptive sensor fusion and client-level personalization across **8 participants** performing **12** meeting-related activities under heterogeneous real-world conditions.

## PUBLICATIONS & ACHIEVEMENTS

- FedMeet**: Personalized Federated Multi-Sensor Fusion for Privacy-Preserving Human Activity Recognition — **27th ACM ICDCN 2026**, peer-reviewed international conference on distributed systems and networking.
- Codeforces Pupil** (Rating: 1100+) — consistent participation in algorithm and data structure contests.
- Qualified **JEE Advanced 2022** (top 2.5% of 1.5M candidates) and **JEE Main 2022**.

## LEADERSHIP

- Core Team, DevLup Labs & Programming Club** — mentored **20+** developers; **50+** active members.
- Tech Head, Ignus & Akshar Festivals** — led **8-member** backend team; **5,000+** attendees, **99.9%** uptime.